H∞ Synthesis

From an energy budget to an implementable controller

An H_∞ synthesis routine does not guess controller coefficients until a frequency plot looks good. It answers a feasibility question: **can any internally stabilizing controller keep the worst-case disturbance-to-performance gain below γ?**

The main difficulty is not solving a quadratic equation. It is understanding why an energy inequality leads to that equation, why output feedback needs **two** equations, and why solving them separately is still not enough.

> Read sections 1–7 first. The route is:
>
> signals ⟶ energy certificate ⟶ state-feedback design ⟶ estimation ⟶ compatibility ⟶ an actual controller.
>
> Section 8 is the numerical workflow; section 9 connects to Youla. Section 10 is a separate guide to all eleven book exercises, with advanced proof outlines labelled.

1 Decide which signals the controller must protect

Use the book's generalized plant, with **positive interconnection** $u = Ky$:

$$\dot{x} = Ax + B_1 w + B_2 u, \quad z = C_1 x + D_{12} u, \quad y = C_2 x + D_{21} w. \tag{1}$$

The sign of corrective feedback is inside K; there is no extra minus sign at the connection. For these first formulas $D_{11} = D_{22} = 0$. General feedthrough is discussed later.

w	Exogenous inputs: disturbances, references, and measurement noise.
u	Actuator commands chosen by the controller.
y	The measurements the controller can actually use.
z	The quantities being penalized, such as weighted error and control effort.

Let $T_{zw} = P \star K$ denote the closed-loop transfer from w to z. For an internally stable finite-dimensional LTI system, with **zero initial state**,

$$\|T_{zw}\|_\infty = \sup_{w \in L_2, w \neq 0} \frac{\|z\|_2}{\|w\|_2} = \sup_{\omega \in \mathbb{R}} \sigma_{\max}(T_{zw}(i\omega)). \tag{2}$$

Here $\|v\|_2^2 = \int_0^\infty v(t)^T v(t)\, \mathrm{d}t$. Thus a bound γ limits energy amplification, not a particular step-response peak, and not the probability of an error. Units and relative priorities belong in the weights already absorbed into P.

The two requirements are distinct: all internal modes must decay, **and** the transfer gain must be below γ. A stable-looking transfer after cancelling a hidden unstable mode does not satisfy the first requirement.

2 Turn a frequency bound into an energy certificate

First freeze the controller, or forget it entirely. Suppose a fixed system is

$$\dot{x} = A_c x + B_c w, \quad z = C_c x, \tag{3}$$

and ask whether it is stable with $\|T_{zw}\|_\infty < \gamma$. Try to account for its stored energy using $V(x) = x^T \Pi x$, $\Pi = \Pi^T > 0$. Require

$$\dot{V} + z^T z - \gamma^2 w^T w < 0 \tag{4}$$

for every nonzero pair (x, w). Expanding gives the matrix test

$$\begin{pmatrix} A_c^T\Pi + \Pi A_c + C_c^T C_c & \Pi B_c \\ B_c^T \Pi & -\gamma^2 I \end{pmatrix} < 0. \tag{5}$$

At $w = 0$ it implies $A_c^T\Pi + \Pi A_c < 0$, hence A_c is Hurwitz. Integrating from $x(0) = 0$ gives $\|z\|_2^2 < \gamma^2\|w\|_2^2$ for nonzero finite-energy inputs. Because the matrix inequality has a strict finite-dimensional margin, it also holds for a slightly smaller positive γ'; this supplies a **uniform** norm gap below γ.

The **strict Bounded Real Lemma** says the converse holds too: stable gain below γ admits such a positive quadratic certificate. Expansion proves sufficiency; necessity is the nontrivial theorem (book §§7.2–7.3), not an assumption about a guessed storage.

The lower-right block is negative definite. Its Schur complement is

$$A_c^T\Pi + \Pi A_c + C_c^T C_c + \gamma^{-2}\Pi B_c B_c^T \Pi < 0. \tag{6}$$

This is a **Riccati inequality**: the last term is quadratic in the unknown Π. No controller has been designed yet. We have only changed the language of the test.

2.1 A complete scalar check

Take $\dot{x} = -3x + w$, $z = 2x$. Its exact gain is

$$\left\| \frac{2}{s+3} \right\|_\infty = \frac{2}{3}. \tag{7}$$

With $\Pi = p > 0$, Equation 6 becomes $-6p + 4 + \frac{p^2}{\gamma^2} < 0$. At $\gamma = 0.8$, choose $p = 1$: the left side is -0.4375, so the certificate works. At $\gamma = 0.6$, even the equality has no real roots: its discriminant is $36 - \frac{16}{0.6^2} < 0$. That demand is impossible, agreeing with the exact gain.

At $\gamma = 0.8$, the equality's roots are $p_- = 0.858680$ and $p_+ = 2.981320$. Both solve the same algebraic equation, but

$$-3 + \frac{p_-}{0.8^2} < 0, \quad -3 + \frac{p_+}{0.8^2} > 0. \tag{8}$$

The first is the **stabilizing Riccati root**. "Stabilizing" refers to this modified matrix, not merely the original $A_c = -3$. Root selection is part of the problem.

> The norm bound became a certificate search. It did not become "solve any quadratic equation and accept any real root."

3 Design state feedback by completing two squares

Now let the controller see the full state and choose $u = Fx$. To keep the algebra visible, assume the normalized control penalty

$$D_{12}^T D_{12} = I, \quad D_{12}^T C_1 = 0. \tag{9}$$

Consequently $z^T z = x^T C_1^T C_1 x + u^T u$: performance charges both state error and control effort, with no cross term. This normalization is not automatic for every plant.

Use storage $V = x^T Y x$, with $Y = Y^T > 0$. Before substituting a feedback law,

$$\dot{V} + z^T z - \gamma^2 w^T w = x^{T(A^T Y + YA + C_1^T C_1)} x + 2x^T Y B_2 u + u^T u + 2x^T Y B_1 w - \gamma^2 w^T w. \tag{10}$$

Complete the control and disturbance squares:

$$\dot{V} + z^T z - \gamma^2 w^T w = x^T \mathcal{R}_{\gamma(Y)} x + \left\| u + B_2^T Y x \right\|^2 - \gamma^2 \left\| w - \gamma^{-2} B_1^T Y x \right\|^2, \tag{11}$$

where

$$\mathcal{R}_{\gamma(Y)} := A^T Y + YA + C_1^T C_1 + Y(\gamma^{-2} B_1 B_1^T - B_2 B_2^T) Y. \tag{12}$$

Read the signs. The controller can eliminate the **positive** square by choosing

$$u = Fx, \quad F = -B_2^T Y. \tag{13}$$

The disturbance maximizes the expression by eliminating the **negative** square. This explains both the negative control term and the positive disturbance term in the Riccati equation. They are not interchangeable signs to memorize.

If $Y > 0$ and $\mathcal{R}_{\gamma(Y)} < 0$, the chosen F gives a strict dissipation certificate and stabilizes the loop. This strict-inequality formulation is an existence test for state-feedback synthesis, not just a sufficient recipe. For necessity, apply the Bounded Real Lemma to any successful F; its Riccati expression is

$$\mathcal{R}_{\gamma(Y)} + \left(F + B_2^T Y\right)^{T\left(F + B_2^T Y\right)} < 0, \tag{14}$$

so dropping the nonnegative square gives $\mathcal{R}_{\gamma(Y)} < 0$.

3.1 Why use an equality instead?

Numerical Riccati algorithms compute a special solution of $\mathcal{R}_{\gamma(Y)} = 0$ rather than searching the whole strict inequality. Under the book's normalized hypotheses, stabilizability of (A, B_2), and its no-imaginary-axis-unobservable-mode condition, the exact state-feedback test is the existence of $Y \geq 0$ satisfying that equality and

$$A + (\gamma^{-2} B_1 B_1^T - B_2 B_2^T) Y \ \text{ is Hurwitz.} \tag{15}$$

Then Equation 13 works. This is a theorem (book exercise 7), not a consequence of the square identity alone. The identity with equality gives a non-strict energy bound; the stabilizing branch and hypotheses supply stability and the strict norm bound.

4 What a Riccati solver actually selects

For a generic equation

$$A^TY + YA + YRY + Q = 0, \quad R = R^T, \quad Q = Q^T, \tag{16}$$

define the Hamiltonian matrix

$$\mathcal{H} = \begin{pmatrix} A & R \\ -Q & -A^T \end{pmatrix}. \tag{17}$$

The apparently nonlinear equation is equivalent to the invariant-subspace identity

$$\mathcal{H}\begin{pmatrix} I \\ Y \end{pmatrix} = \begin{pmatrix} I \\ Y \end{pmatrix}(A + RY). \tag{18}$$

Indeed, its top row is $A + RY$; its bottom row is $-Q - A^TY = Y(A + RY)$.

If the stable invariant subspace can be written as $\begin{pmatrix} U \\ V \end{pmatrix}$ with U invertible, then $Y = VU^{-1}$. Its associated matrix $A + RY$ is Hurwitz because the subspace was selected from the left-half-plane eigenvalues. In computation use an ordered Schur method, solve for the quotient, and check the residual; avoid explicit inversion.

For the control equation, $R = \gamma^{-2}B_1B_1^T - B_2B_2^T$ can be **indefinite**. Thus these are separate checks:

1. no Hamiltonian eigenvalues on the imaginary axis;
2. the selected subspace has an invertible, numerically usable top block;
3. the recovered solution is symmetric, positive semidefinite, and has a small residual;
4. the specified Riccati stability matrix is Hurwitz.

The first check alone is not a universal theorem for indefinite Riccati equations. For example, $A = 1$, $R = Q = 0$ gives $\mathcal{H} = \mathrm{diag}(1, -1)$: there are no imaginary eigenvalues, but the stable subspace has $U = 0$, and $A + RY = 1$ cannot be stabilized by any Riccati root. The book's early equivalences impose extra sign and controllability assumptions; do not carry them silently into a different equation.

4.1 Why does the book sometimes ask for an anti-stabilizing root?

Our Y is the control-cost coordinate used as Y_∞ in Theorem 7.13. Earlier, Theorem 7.9 uses the inverse coordinate $W = Y^{-1}$ when $Y > 0$. Multiplying the control ARE on both sides by W gives

$$AW + WA^T + \gamma^{-2}B_1B_1^T - B_2B_2^T + WC_1^TC_1W = 0. \tag{19}$$

Its branch matrix is $A + WC_1^TC_1$, and the ARE implies

$$A + WC_1^TC_1 = -W\big(A + (\gamma^{-2}B_1B_1^T - B_2B_2^T)Y\big)^TW^{-1}. \tag{20}$$

The eigenvalues are negated. A stabilizing Y therefore corresponds to an **anti-stabilizing** W, with the same physical gain $F = -B_2^TW^{-1}$. Nothing physically changed stability; only the certificate coordinate changed. If Y is singular, this inverse coordinate is unavailable, another reason to prefer the semidefinite formulation in the final theorem.

5 Estimation is the dual problem — but output feedback needs more

If x is not measured, try an observer for

$$\dot{x} = Ax + B_1 w, \quad z = C_1 x, \quad y = C_2 x + D_{21} w. \tag{21}$$

Use the book's sign convention

$$\dot{\hat{x}} = A\hat{x} + L(C_2\hat{x} - y), \quad \hat{z} = C_1\hat{x}. \tag{22}$$

With $\xi = x - \hat{x}$, subtract the equations:

$$\dot{\xi} = (A + LC_2)\xi + (B_1 + LD_{21})w, \quad e = C_1\xi. \tag{23}$$

Thus observer design asks for stable error dynamics and gain from w to **estimation error** e below γ. Transpose that transfer matrix. It becomes a state-feedback problem with the substitutions

$$(A, B_1, B_2, C_1, D_{12}) \to (A^T, C_1^T, C_2^T, B_1^T, D_{21}^T). \tag{24}$$

The dual normalization is $B_1 D_{21}^T = 0$, $D_{21} D_{21}^T = I$. The corresponding filter ARE is

$$AX + XA^T + B_1 B_1^T + X(\gamma^{-2} C_1^T C_1 - C_2^T C_2)X = 0, \tag{25}$$

with $X \geq 0$ and $A + X(\gamma^{-2} C_1^T C_1 - C_2^T C_2)$ Hurwitz. The observer gain is $L = -XC_2^T$. As on the control side, appropriate detectability and regularity hypotheses are needed for the ARE characterization; this is not a formula for arbitrary plant data.

5.1 Why not just combine the two independently?

In LQG, the control and filtering Riccati equations can be solved independently. At a finite H_∞ bound, the same adversarial input affects what must be controlled **and** what can be inferred. Estimation and control must fit within one performance budget. Two individually successful Riccati solutions need not be compatible.

The extra condition is

$$\rho(XY) < \gamma^2, \tag{26}$$

where ρ is spectral radius. Since $X, Y \geq 0$, the nonzero eigenvalues of XY are those of the symmetric positive-semidefinite $X^{\frac{1}{2}} Y X^{\frac{1}{2}}$. Hence the condition can also be read as

$$I - \gamma^{-2} X^{\frac{1}{2}} Y X^{\frac{1}{2}} > 0. \tag{27}$$

It is a strict compatibility margin, not a third Riccati equation and not a replacement for either equation's stability test. It makes

$$Z = (I - \gamma^{-2} XY)^{-1} \tag{28}$$

well defined. Z need not be symmetric. Keep its matrix ordering exactly as written.

> State feedback answers "can I control with perfect information?" Filtering answers "can I estimate what matters?" The coupling test asks whether those two solutions can be assembled at the **same** bound.

6 Assemble the central controller

Here is the normalized two-Riccati result (book Theorem 7.13), in one place. Assume

1. $D_{11} = D_{22} = 0$, $D_{12}^T D_{12} = I$, $D_{12}^T C_1 = 0$, $D_{21} D_{21}^T = I$, $B_1 D_{21}^T = 0$;
2. (A, B_2) is stabilizable and (A, C_2) is detectable;
3. (A, B_1) has no uncontrollable modes on the imaginary axis and (A, C_1) has no unobservable modes there.

Then a proper LTI controller internally stabilizing P with $\|P \star K\|_\infty < \gamma$ exists **if and only if** the filter ARE Equation 25 and the control ARE $\mathcal{R}_{\gamma(Y)} = 0$ have their specified stabilizing solutions $X, Y \geq 0$ and Equation 26 holds. With Z from Equation 28, one such controller is

$$\dot{x}_K = A_K x_K + B_K y, \quad u = C_K x_K, \tag{29}$$

$$A_K = A - ZXC_2^T C_2 + (\gamma^{-2} B_1 B_1^T - B_2 B_2^T) Y, \tag{30}$$

$$B_K = ZXC_2^T, \quad C_K = -B_2^T Y, \quad D_K = 0. \tag{31}$$

The controller has a realization with the plant's state dimension. A transfer-function reduction may lower that order; there is no need to promise it is always minimal.

6.1 Read the formula as an observer with a worst-case correction

Set $F = -B_2^T Y$, $L = -XC_2^T$, and $\hat{A} = A + \gamma^{-2} B_1 B_1^T Y$. The same controller is

$$\dot{\hat{x}} = \hat{A}\hat{x} + B_2 u + ZL(C_2 \hat{x} - y), \quad u = F\hat{x}. \tag{32}$$

It resembles the ordinary observer/state-feedback construction, but has two important changes: the worst-case drift correction in $\hat{A}$, and the coupling factor Z in the injection gain. Dropping either generally gives a different controller with no guarantee at the requested level. This is **not** the ordinary independent LQG separation.

As $\gamma \to \infty$, the adversarial terms vanish and $Z \to I$, provided the usual stabilizing LQG solutions exist. The equations reduce to the control and filter Riccati equations of LQG. X and Y at finite γ are not automatically stochastic covariances: the derivation was a deterministic worst-case energy argument.

6.2 Check the loop you actually connected

For $D_K = D_{11} = D_{22} = 0$, the complete closed-loop realization is

$$A_c = \begin{pmatrix} A & B_2 C_K \\ B_K C_2 & A_K \end{pmatrix}, \quad B_c = \begin{pmatrix} B_1 \\ B_K D_{21} \end{pmatrix}, \tag{33}$$

$$C_c = \begin{pmatrix} C_1 & D_{12} C_K \end{pmatrix}, \quad D_c = 0. \tag{34}$$

Check **all** eigenvalues of A_c before any pole–zero cancellation, and compute the H_∞ norm of this realization. A Riccati residual is not a substitute for checking that the correct signs, signal ordering, and controller were implemented.

7 One plant, from hand calculation to a controller

Consider a scalar state with unit process disturbance d and measurement noise v:

$$\dot{x} = -x + d + u, \quad y = x + v, \quad z = \begin{pmatrix} x \\ u \end{pmatrix}, \quad w = \begin{pmatrix} v \\ d \end{pmatrix}. \tag{35}$$

We want small state excursions without unlimited actuator effort, using only the noisy measurement. In Equation 1 this means

$$A = -1, \quad B_1 = \begin{pmatrix} 0 & 1 \end{pmatrix}, \quad B_2 = 1, \quad C_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad D_{12} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad C_2 = 1, \quad D_{21} = \begin{pmatrix} 1 & 0 \end{pmatrix}. \tag{36}$$

The normalized assumptions hold. Both Riccati equations reduce to

$$-2a + (\gamma^{-2} - 1)a^2 + 1 = 0. \tag{37}$$

The stabilizing root, written without a removable singularity at $\gamma = 1$, is

$$X = Y = a(\gamma) = \frac{1}{1 + \sqrt{2 - \gamma^{-2}}}, \quad \gamma > \frac{1}{\sqrt{2}}. \tag{38}$$

Its branch eigenvalue is $-\sqrt{2 - \gamma^{-2}} < 0$. This verifies the root selection, not just substitution into the quadratic.

7.1 Three requests with three different answers

Requested γ	Why it succeeds or fails
0.70	No real stabilizing Riccati root: $2 - \gamma^{-2} < 0$.
0.72	Both AREs have stabilizing roots, but $\frac{a^2}{\gamma^2} > 1$: coupling fails.
0.75	Both roots pass, $a = 0.679623$, and $\frac{a^2}{\gamma^2} = 0.821133 < 1$.

The output-feedback threshold is therefore **not** the point where either individual Riccati equation first becomes solvable. It is where $a(\gamma) = \gamma$. Since

$$\gamma + \sqrt{2\gamma^2 - 1} = 1, \tag{39}$$

squaring within the relevant interval gives

$$\gamma^2 + 2\gamma - 2 = 0, \quad \gamma_* = \sqrt{3} - 1 = 0.7320508076\ldots. \tag{40}$$

The feasible strict bounds are $\gamma > \gamma_*$. This is an infimum over controllers; the feasibility theorem makes no promise of a finite controller at the endpoint.

7.2 Write down the controller at gamma = 0.75

Substitution into Equation 31 gives

$$A_K = -4.270995, \quad B_K = 3.799590, \quad C_K = -0.679623, \quad D_K = 0. \tag{41}$$

Thus $K(s) = C_K \frac{B_K}{s - A_K}$ has a **negative** transfer gain. This is the corrective sign inside $u = Ky$, not a reason to insert a second minus sign in the feedback command. The physical closed-loop norm is $0.749283 < 0.75$. The requested certificate level and the achieved norm are close in this case, but are not the same number.

7.3 What changes as we demand better performance?

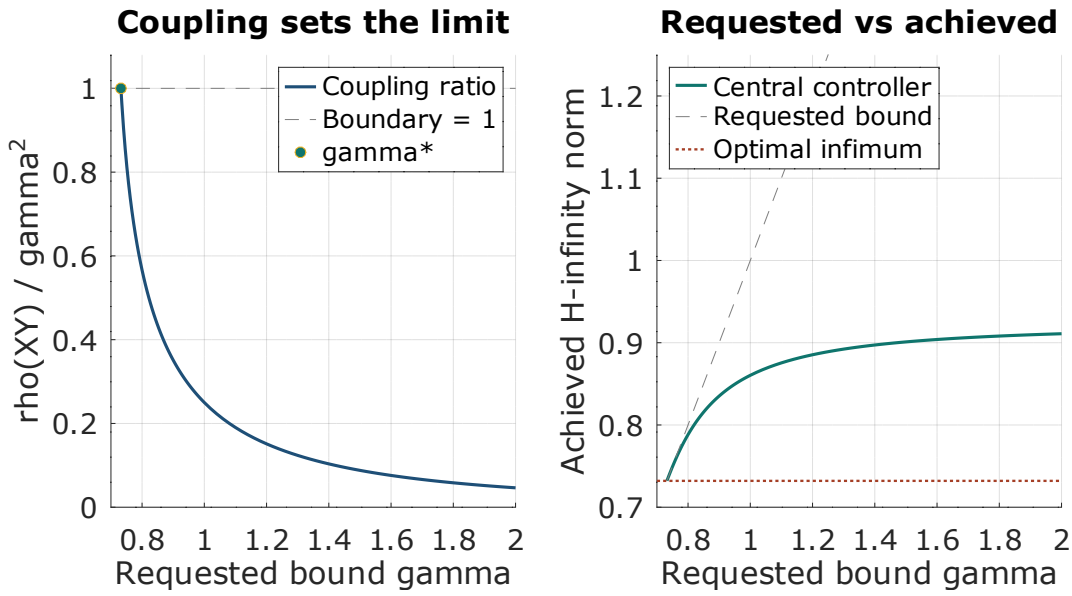


Figure 1: Computed from the physical plant/controller interconnection. Left: compatibility becomes tight at γ_*. Right: the achieved norm stays below each feasible requested bound but above the optimal infimum, approaching γ_* from above.

γ	a	$\frac{a^2}{\gamma^2}$	A_K	Actual norm
0.7321	0.731867	0.999365	−1152.160	0.732100
0.75	0.679623	0.821133	−4.270995	0.749283
1.00	0.500000	0.250000	−1.666667	0.860380
1.20	0.466721	0.151270	−1.692514	0.885374
2.00	0.430501	0.046333	−1.774292	0.910936
∞	0.414214	0	−1.828427	0.921519

The near-endpoint norm rounds to the requested bound in the table; the unrounded computation is still strictly below it. As $\gamma \to \gamma_*^+$, the denominator of $Z = \left(1 - \frac{a^2}{\gamma^2}\right)^{-1}$ tends to zero. Both A_K and B_K become large in magnitude. The controller acquires fast dynamics; asking for nearly optimal performance can be numerically and physically expensive. These numbers demonstrate that issue for the central family, not a universal proof that every plant's optimum must be high-gain.

If the exact state were available, $u = fx$ gives

$$\|T_{zd}\|_\infty = \frac{\sqrt{1+f^2}}{1-f}, \quad f < 1. \tag{42}$$

Its squared value minus $\frac{1}{2}$ is $\frac{(f+1)^2}{2(1-f)^2}$, so its minimum is exactly $\frac{1}{\sqrt{2}}$ at $f = -1$. The noisy-output problem has the larger threshold $\sqrt{3} - 1$. This makes the extra cost of limited information concrete.

At $\gamma = \infty$, $a = \sqrt{2} - 1$ and the central controller is the LQG limit. Its H_∞ norm is 0.921519, while its H_2 norm is 0.696621. They measure different things. The latter is computed by the Lyapunov-based norm routine, not by a truncated frequency integral that misses energy near zero and infinity.

8 Use the theorem as a design workflow, not as a black box

The feasible set of requested bounds is upward closed: if one controller achieves γ_0, it achieves every larger bound too. This is why bisection is justified.

1. Build P from the actual signals and weights. Check dimensions and the feedback sign.
2. Find an infeasible lower bound and a feasible upper bound. For each midpoint, solve both AREs on the specified branches; check residuals, semidefiniteness, and branch stability.
3. Reject the midpoint if either Riccati test fails or $\rho(XY) \geq \gamma^2$.
4. If all tests pass, construct K, check the complete closed-loop poles and actual norm, and lower the feasible upper bound. Stop with a tolerance, not at a singular endpoint.
5. Inspect actuator gains, sensitivity, and time responses as engineering checks in addition to the certificate. A weighted energy bound alone does not guarantee no saturation.

In exact arithmetic these tests decide feasibility under the theorem's hypotheses. Near the boundary a solver can have numerical difficulty. Report that separately from a mathematical infeasibility certificate, and retain a useful safety margin.

8.1 General feedthrough and non-normalized data

For a fixed closed-loop system $z = C_c x + D_c w$, the strict Bounded Real LMI is

$$\begin{pmatrix} A_c^T\Pi + \Pi A_c + C_c^T C_c & \Pi B_c + C_c^T D_c \\ B_c^T\Pi + D_c^T C_c & D_c^T D_c - \gamma^2 I \end{pmatrix} < 0, \quad \Pi > 0. \tag{43}$$

It includes the necessary high-frequency condition $\|D_c\| < \gamma$. Do not use the zero-feedthrough formulas unchanged on arbitrary plant matrices. General regular synthesis routines account for feedthrough, cross terms, and scalings. Their standard assumptions include stabilizability/detectability, full column rank of D_{12}, full row rank of D_{21}, and the appropriate plant pencils having full rank at every imaginary frequency (book §7.9).

One transparent regularization adds channels rather than hiding their effect:

$$\dot{x} = Ax + B_1 w + \varepsilon w_1 + B_2 u, \tag{44}$$

$$z_{\text{aug}} = \begin{pmatrix} C_1 x + D_{11} w + D_{12} u \\ \varepsilon x \\ \varepsilon u \end{pmatrix}, \tag{45}$$

$$y = C_2 x + D_{21} w + \varepsilon w_2 + D_{22} u. \tag{46}$$

The controller still sees the same measurement dimension and drives the same actuators. The full autonomous plant/controller matrix is unchanged: new external inputs and penalized outputs do not add internal states. Thus internal stability is unchanged. For each fixed stabilizing K, the augmented norm is at least the original one and converges to it as $\varepsilon \to 0$. A strict bound for the original problem survives for sufficiently small ε; a bound for the augmented problem also protects the original. An arbitrary finite ε nevertheless changes the optimization problem.

9 Two deeper readings: a game and a family of controllers

9.1 The energy identity is also a zero-sum game

With $\mathcal{R}_{\gamma(Y)} = 0$, integrate Equation 11 from $x(0) = \xi$:

$$x(T)^T Y x(T) - \xi^T Y \xi + \int_0^T \left(\|z\|^2 - \gamma^2 \|w\|^2\right) \mathrm{d}t \tag{47}$$

$$= \int_0^T \left\|u + B_2^T Y x\right\|^2 \mathrm{d}t - \gamma^2 \int_0^T \left\|w - \gamma^{-2} B_1^T Y x\right\|^2 \mathrm{d}t. \tag{48}$$

The controller minimizes the payoff and the disturbance maximizes it. Their candidate strategies are $u_* = -B_2^T Y x$ and $w_* = \gamma^{-2} B_1^T Y x$. Together they give Equation 15, so the saddle trajectory decays and both inputs have finite energy. For this pair the infinite-horizon payoff equals $\xi^T Y \xi$.

For arbitrary finite-energy w, the stabilizing u_* guarantees payoff at most that value. Under the book's game hypotheses, including observability of (A, C_1), the converse minimax/maximin equalities require the admissibility arguments of Theorems 7.20, 7.26, and 7.27. Opposite squares alone do not justify interchanging an infimum and a supremum for arbitrary strategies.

The inner-system construction in §7.11 repackages this balance as a lossless channel: an inner stable transfer preserves L_2 energy. It allows the performance condition to be transferred to a free stable bounded parameter. Its role is an exact change of interconnection coordinates, not an approximation or an extra physical component.

9.2 Central means the zero member of a whole family

Let $F = -B_2^T Y$, $L = -X C_2^T$, and let A_K be as in Equation 31. Define

$$K_{\mathrm{par}} = \begin{pmatrix} A_K & -ZL & ZB_2 \\ F & 0 & I \\ -C_2 & I & 0 \end{pmatrix}. \tag{49}$$

The book's Theorem 7.40 states that all internally stabilizing controllers meeting the strict bound have the form

$$K = K_{\mathrm{par}} \star Q_\gamma, \quad Q_\gamma \in \mathcal{RH}_\infty, \quad \left\|Q_\gamma\right\|_\infty < \gamma. \tag{50}$$

Here $\mathcal{RH}_\infty$ means real-rational, proper, with all poles strictly in the left half-plane. The lower LFT means that the second output of K_{par} is sent through Q_γ and returned at its second input. Its lower-right feedthrough is zero, so the connection is well posed for proper Q_γ. Setting $Q_\gamma = 0$ recovers the central controller. The realized order is at most the plant order plus the parameter order.

> Do not confuse this norm-ball parameter with chapter 9's unrestricted Youla parameter. Here the parameter makes the **performance constraint** simple; the physical closed-loop map is generally fractional, not affine, in it. Chapter 9 uses different coordinates to make the stable closed-loop map affine.

This is the handoff to chapter 8: each fixed-scaling controller step is an H_∞ synthesis problem. The two-Riccati theorem does not itself choose uncertainty scalings or solve the joint structured robust-performance problem.

10 Exercise guide: mechanisms before technical details

This guide covers the eleven exercises on pp. 211–216. Exercises 1–5, 8, 10, and 11 have the central algebra worked out below; exercises 6, 7, and the endpoint part of 9 include explicitly identified proof outlines. The technical existence and boundary arguments are not presented as full proofs. First understand the design route above.

10.1 Exercise 1 — symmetry, reality, and uniqueness of the stabilizing root

Let X solve $A^TX + XA + XRX + Q = 0$, initially allowing complex nonsymmetric X, and let $M = A + RX$ be Hurwitz. Subtract the transposed equation. For $\Delta = X - X^T$,

$$M^T\Delta + \Delta M = 0. \tag{51}$$

Its only solution is zero: $e^{M^Tt}\Delta e^{Mt}$ is constant and tends to zero. Thus $X = X^T$. If X_1, X_2 are symmetric stabilizing solutions, subtraction gives

$$(A + RX_1)^{T(X_1 - X_2)} + (X_1 - X_2)(A + RX_2) = 0. \tag{52}$$

The same exponential argument proves uniqueness, since both coefficient matrices are Hurwitz. The complex conjugate $|X|$ is another symmetric stabilizing solution for real data, so uniqueness implies $|X| = X$. No separate claim about a real Hamiltonian subspace is needed.

10.2 Exercise 2 — detect hidden unstable modes with the right rank tests

The lemma concerns every λ in the **closed right half-plane**, not just imaginary frequencies. Its two tests are full column rank of

$$\begin{pmatrix} A - \lambda I & B_2 \\ C_1 & D_{12} \end{pmatrix} \tag{53}$$

and full row rank of

$$\begin{pmatrix} A - \lambda I & B_1 \\ C_2 & D_{21} \end{pmatrix}. \tag{54}$$

To see the first test's role, suppose the complete closed-loop matrix has an unstable or marginal eigenvector $\begin{pmatrix} x \\ \eta \end{pmatrix}$ invisible at z. With $u = C_K\eta + D_KC_2x$, the plant equations give $(A - \lambda I)x + B_2u = 0$ and $C_1x + D_{12}u = 0$. Full column rank forces $x = u = 0$. Then $A_K\eta = \lambda\eta$ and $C_K\eta = 0$. A detectable controller realization excludes such a nonzero η. The row-rank test, applied dually, excludes modes unreachable from w if the controller realization is stabilizable. Therefore no closed-right-half-plane mode is hidden from the $w \to z$ transfer. Minimal controller realizations satisfy both requirements.

With $D_{12}^+ = (D_{12}^TD_{12})^{-1}D_{12}^T$, eliminate u from the first nullspace equations:

$$(A - B_2D_{12}^+C_1 - \lambda I)x = 0, \quad (I - D_{12}D_{12}^+)C_1x = 0. \tag{55}$$

That is exactly the PBH detectability test for the pair stated in the exercise. Transposing gives the stabilizability test with $D_{21}^+ = D_{21}^{T(D_{21}D_{21}^T)^{-1}}$. Do not extend this conclusion to a controller with an arbitrary hidden unstable state.

10.3 Exercises 3–5 — why the LQ value is quadratic and obeys an ARE

Let $\dot{x} = Ax + Bu$, $x(0) = \xi$, with (A, B) stabilizable, and let

$$q(x,u) = \begin{pmatrix} x \\ u \end{pmatrix}^T \begin{pmatrix} Q & S \\ S^T & R \end{pmatrix} \begin{pmatrix} x \\ u \end{pmatrix}, \quad Q = Q^T, \quad R = R^T \geq 0. \tag{56}$$

Write $J(\xi, u) = \int_0^\infty q \, \mathrm{d}t$ for admissible $u, x \in L_2$, and assume $v(\xi) = \inf_u J(\xi, u) > -\infty$ for every ξ.

3(a–c): scaling and concatenation. The zero trajectory gives $v(0) \leq 0$. Any negative cost from zero could be multiplied by arbitrarily large squared amplitude, contradicting finiteness, so $v(0) = 0$. Concatenate a finite trajectory with an ε-optimal tail:

$$v(x(t_1)) \leq \int_{t_1}^{t_2} q \, \mathrm{d}t + v(x(t_2)). \tag{57}$$

Taking the infimum over the first segment and using near-optimal complete trajectories gives equality in the dynamic-programming principle. No optimizer needs to exist.

3(d): the parallelogram mechanism. Adding/subtracting two admissible trajectories and using quadraticity gives

$$v(\xi + \eta) + v(\xi - \eta) \leq 2v(\xi) + 2v(\eta). \tag{58}$$

Apply the same construction to near-optimal trajectories from $\xi + \eta$ and $\xi - \eta$, then halve their sum/difference, to obtain the reverse inequality. Scaling also gives $v(a\xi) = a^2 v(\xi)$. A fixed stabilizing feedback gives the upper bound $v(\xi) \leq c\|\xi\|^2$. The nonnegative parallelogram form $c\|\xi\|^2 - v(\xi)$ is bounded on a finite-dimensional unit ball by its values on a basis, supplying the required local boundedness. Polarization then gives $v(\xi) = \xi^T K \xi$, $K = K^T$.

4: differentiate the optimality inequality. The short-time expansion is

$$0 \leq q(x,u) + \frac{\mathrm{d}}{\mathrm{d}}t(x^T K x). \tag{59}$$

Hence the exercise's block LMI holds. Notice the **plus** sign before the derivative. For any other symmetric LMI solution X, integrate its inequality to obtain

$$\xi^T X \xi \leq \int_0^\infty q \, \mathrm{d}t, \tag{60}$$

since $x(t) \to 0$ for $x, \dot{x} \in L_2$. Infimizing gives $X \leq K$. Reversing the derivative inequality would give the wrong conclusion.

5(a–b): remove cross terms and take a Schur complement. With $R > 0$, put

$$\hat{u} = R^{\frac{1}{2}}(u + R^{-1}S^T x), \quad \tilde{Q} = Q - SR^{-1}S^T. \tag{61}$$

Then $q = x^T \tilde{Q} x + \|\hat{u}\|^2$ and $\dot{x} = (A - BR^{-1}S^T)x + BR^{-\frac{1}{2}}\hat{u}$. This is a bijection of admissible pairs and costs. For normalized $S = 0, R = I$, the block LMI is equivalent to

$$g(X) := A^T X + XA - XBB^T X + Q \geq 0. \tag{62}$$

5(c–e): the equality comes from zero residual value. For the finite-horizon Riccati solution $P(T) = 0$, the correct square identity is

$$\int_0^T (\|Cx\|^2 + \|u\|^2) \, \mathrm{d}t = \xi^T P(0) \xi + \int_0^T \left\|u + B^T P x\right\|^2 \mathrm{d}t. \tag{63}$$

It proves the minimum and its feedback. If that minimum is zero for every initial state, nonnegativity forces $C\xi = 0$ for every ξ, hence $C = 0$. Finally factor $g(K) = C^T C$ and shift to $\hat{u} = u + B^T K x$. The original infinite-horizon value minus $\xi^T K \xi$ is precisely the infimum of $\int(\|Cx\|^2 + \|\hat{u}\|^2)$. It is zero. Every finite-horizon minimum is bounded by this nonnegative infinite-horizon infimum, so it too is zero. Thus $C = 0$ and $g(K) = 0$: the value matrix satisfies the ARE.

10.4 Exercise 6 — the strong solution is the largest ARI solution (proof outline)

"Strong" permits imaginary-axis eigenvalues of $M = A - BB^T X_s$; "stabilizing" does not. For $\Delta = X - X_s$, subtraction gives

$$g(X) = M^T \Delta + \Delta M - \Delta B B^T \Delta \geq 0. \tag{64}$$

If M is strictly Hurwitz, the Lyapunov integral immediately yields $\Delta \leq 0$. For a merely strong solution that argument is not valid: the integral may not converge. One must handle the marginal modes using stabilizability, or pass through a vanishing positive cost regularization. This is the technical step in the book's kernel/modal hint.

A useful alternative proof route is the stabilizing LQ value. Once a strong X_s is known, completion of squares gives cost

$$\xi^T X_s \xi + \int_0^\infty \left\| u + B^T X_s x \right\|^2 \mathrm{d}t. \tag{65}$$

The stabilizable marginal modes of M can be stabilized with arbitrarily small residual energy, so the infimum is $\xi^T X_s \xi$. Exercise 4(b) then bounds every ARI solution by X_s. Conversely the maximal ARE solution must be strong: a right-half-plane modal block would allow an upward ARE perturbation. Thus two strong solutions bound each other and coincide.

For existence, start from an ARI solution X_0, shift the dynamics to $A - BB^T X_0$, and use nonnegative running state cost $g(X_0)$. The stabilizing LQ value gives a strong residual solution $Z \geq 0$; then $X_s = X_0 + Z$ solves the original ARE. Constructing that LQ value as a limit and proving the marginal-mode assertions are the parts of this outline not reproved here. If $Q \geq 0$, $X_0 = 0$ is available and maximality gives $X_s \geq 0$.

10.5 Exercise 7 — weak observability hypotheses (proof outline)

The desired control ARE and gain are Equation 12 and Equation 13. Scaling the disturbance by $\frac{1}{\gamma}$ puts the problem at unit bound. If the relevant control-cost solution is positive definite, the inverse-coordinate argument of section 4 reduces it to Theorem 7.9. But positivity may fail when stable modes are invisible to C_1. Do not invert a possibly singular solution.

The correct reduction first separates the **stable unobservable** part. In coordinates isolating that invariant part one has

$$A = \begin{pmatrix} A_o & 0 \\ A_{21} & A_s \end{pmatrix}, \quad C_1 = \begin{pmatrix} C_o & 0 \end{pmatrix}, \tag{66}$$

where A_s is Hurwitz for that removed part. Solve on the remaining subsystem and extend the cost matrix with a zero block on the removed states. The closed-loop and game stability matrices are triangular with the untouched stable block A_s. This gives a semidefinite solution without an illegal inversion. Any unobservable unstable modes must remain in the Riccati problem, not be discarded. Completing the necessity proof uses the corresponding stabilizing Riccati existence theorem.

10.6 Exercise 8 — general control penalties

Let $R_u = D_{12}^T D_{12} > 0$, $N = C_1^T D_{12}$, and set

$$u = -R_u^{-1} N^T x + R_u^{-\frac{1}{2}} \hat{u}. \tag{67}$$

Then the state penalty is $C_1^T C_1 - N R_u^{-1} N^T$, and applying the normalized construction yields

$$A^T Y + Y A + \gamma^{-2} Y B_1 B_1^T Y + C_1^T C_1 - (Y B_2 + N) R_u^{-1} (B_2^T Y + N^T) = 0, \tag{68}$$

with $F = -R_u^{-1}(B_2^T Y + N^T)$ and branch matrix $A + B_2 F + \gamma^{-2} B_1 B_1^T Y$. The imaginary-axis pencil condition ensures the projected state penalty has no invisible marginal modes, so exercise 7 gives the weaker rank hypothesis asked for. An imaginary-axis invariant zero does **not** make every closed-loop norm infinite; it makes this particular strict Riccati equivalence require care. Stable loops can have zeros on that axis.

10.7 Exercise 9 — the Riccati existence boundary is not always the design boundary

Use the inverse coordinate $W(\gamma)$ from section 4; this is the exercise's $Y_+(\gamma)$. Let $W_0 = W(\infty)$ and $M_0 = A + W_0 C_1^T C_1$, which is anti-Hurwitz. For $E = W_0 - W(\gamma)$, subtract the two inverse AREs:

$$(-M_0)E + E(-M_0)^T + E C_1^T C_1 E + \gamma^{-2} B_1 B_1^T = 0. \tag{69}$$

This is the dual Bounded Real ARE for the **stable** residual system with state matrix $-M_0$. Its stabilizing branch corresponds to the anti-stabilizing branch for W. It identifies the boundary

$$\gamma_{\min} = \left\| C_1 (sI + A + W_0 C_1^T C_1)^{-1} B_1 \right\|_\infty. \tag{70}$$

The printed exercise places W_0 on the other side of $C_1^T C_1$. The subtraction above fixes the matrix order; the two orders need not agree. The executable example checks this with noncommuting 2×2 matrices rather than a scalar case.

Inside the open domain, differentiating the inverse ARE gives

$$M_\gamma \dot{W} + \dot{W} M_\gamma^T = 2\gamma^{-3} B_1 B_1^T, \quad M_\gamma = A + W(\gamma) C_1^T C_1. \tag{71}$$

Since M_γ is anti-Hurwitz,

$$\dot{W} = \int_0^\infty e^{-M_\gamma t} (2\gamma^{-3} B_1 B_1^T) e^{-M_\gamma^T t} \, \mathrm{d}t \geq 0. \tag{72}$$

The Lyapunov operator is invertible there, also giving smoothness.

Endpoint interpretation and proof outline. The inverse ARE exists on $(\gamma_{\min}, \infty)$, but state-feedback feasibility additionally needs $W(\gamma) > 0$. The endpoint symbols in parts (c–e) must be read as appropriate one-sided limits, not as a claim that the branch matrix stays strictly anti-Hurwitz there. If the lower endpoint limit is positive definite, its finite gain can attain the **non-strict** optimal norm. If positivity first becomes possible at a singular limit, the infimal state-feedback gains must become unbounded under the exercise's hypotheses. The proof uses a bounded-subsequence argument plus a separate stabilization of the limiting marginal modes; continuity of eigenvalues alone is not enough. That boundary proof is not reconstructed here.

Our scalar plant illustrates the first case for **state feedback**: $W_0 = 1 + \sqrt{2}$, $M_0 = \sqrt{2}$, $\gamma_{\min} = \frac{1}{\sqrt{2}}$, and the endpoint $W = 1$ gives the finite optimal gain $F = -1$. The output-feedback central family, by contrast, meets a **coupling** boundary at the larger $\sqrt{3} - 1$. These are different obstacles.

10.8 Exercise 10 — the general ARIs and a correct large-storage argument

At bound one let $D_c = (D_{12}^T D_{12})^{-1} D_{12}^T$ and $D_r = D_{21}^{T(D_{21}D_{21}^T)^{-1}}$. Normalize the control penalty as in exercise 8 and normalize the dual measurement penalty. The two **inverse-coordinate** ARIs are

$$A_u Y + Y A_u^T + Y Q_u Y + R_u < 0, \tag{73}$$

$$A_y^T X + X A_y + X Q_y X + R_y < 0, \tag{74}$$

where

$$A_u = A - B_2 D_c C_1, \quad Q_u = C_1^{T(I - D_{12}D_c)} C_1, \tag{75}$$

$$R_u = B_1 B_1^T - B_2 (D_{12}^T D_{12})^{-1} B_2^T, \tag{76}$$

$$A_y = A - B_1 D_r C_2, \quad Q_y = B_1 (I - D_r D_{21}) B_1^T, \tag{77}$$

$$R_y = C_1^T C_1 - C_2^{T(D_{21}D_{21}^T)^{-1}} C_2. \tag{78}$$

Together with $X > 0$, $Y - X^{-1} > 0$, these are the exercise's general two-ARI criterion. Here X, Y are **not** the X_∞, Y_∞ coordinates of Theorem 7.13. The displayed products also correct dimensionally inconsistent transposes in the printed second inequality. Completing squares derives the terms; the two-ARI output-feedback theorem supplies the existence/construction equivalence.

Part (b), fully. Given Hurwitz A and symmetric Q, solve

$$A^T H + H A = -I, \quad A^T X_0 + X_0 A = -Q. \tag{79}$$

Then $H > 0$, and $X = X_0 + tH$ satisfies

$$A^T X + X A + Q = -tI < 0. \tag{80}$$

For sufficiently large $t > 0$, it also satisfies $X > \alpha I$ for any prescribed real α. Adding tI instead is not valid: Hurwitz does not imply $A + A^T < 0$. For example $A = \begin{pmatrix} -1 & 10 \\ 0 & -2 \end{pmatrix}$ is Hurwitz but its symmetric part has a positive eigenvalue. This asymmetric example is checked in the script.

Part (c). If D_{21} is square, $Q_y = 0$. If A_y is Hurwitz, part (b) supplies arbitrarily large positive X satisfying its now-linear ARI. For any fixed $Y > 0$ solving the control ARI, choose $X > Y^{-1}$; then $X^{-1} < Y$, so coupling also holds. Only the control-side inequality remains to be tested.

10.9 Exercise 11 — unstable zeros become interpolation constraints

Use $P = \begin{pmatrix} h & -g \\ 1 & 0 \end{pmatrix}$ so that $P \star f = h - gf$. Since this is a stable feedforward plant, internal stability amounts to $f \in \mathcal{RH}_\infty$. If $e = h - gf$, every right-half-plane zero z_j of g enforces $e(z_j) = h(z_j)$: a stable f cannot cancel that condition with an unstable pole. Conversely a stable proper e satisfying these interpolation conditions makes $f = \frac{h-e}{g}$ stable and proper, because $g(\infty) = 1$ and the only inverse poles are its stated simple right-half-plane zeros.

The strict scalar Nevanlinna–Pick criterion for an error norm below γ is

$$\mathcal{P}_{jk} = \frac{\gamma^2 - h(z_j)\overline{h(z_k)}}{z_j + \overline{z_k}}, \quad \mathcal{P} > 0. \tag{81}$$

At $\gamma = 1$ this is the exercise's matrix. The remaining argument connects it to the Lyapunov equation, rather than asserting that a fixed-controller norm LMI by itself solves model matching.

Choose modal coordinates for the inverse-zero matrix

$$A_0 = A_g - B_g C_g = \operatorname{diag}(z_1, \ldots, z_k), \quad B_g = \mathbb{1}. \tag{82}$$

Simple zeros and minimality allow this normalization; use conjugate modal coordinates for nonreal zeros, with Hermitian products rather than real transposes. The Sylvester equation

$$A_0 U - U A_h = B_g C_h \tag{83}$$

gives row by row $e_j^T U = C_{h(z_j I - A_h)}^{-1}$, hence $e_j^T U B_h = h(z_j)$. The Lyapunov equation for the general bound is

$$A_0 V + V A_0^* = \gamma^2 B_g B_g^* - U B_h B_h^* U^*. \tag{84}$$

In entries this is

$$(z_j + \overline{z_k}) V_{jk} = \gamma^2 - h(z_j)\overline{h(z_k)}, \tag{85}$$

so $V = \mathcal{P}$. Since A_0 is anti-Hurwitz, its Lyapunov solution is unique. Thus the positivity criterion on V in part (b) is exactly the Pick test in part (e). The associated single Lyapunov **inequality** may be written

$$A_0 V + V A_0^* - \left(\gamma^2 B_g B_g^* - U B_h B_h^* U^*\right) < 0, \quad V > 0. \tag{86}$$

It is feasible iff the equality solution is positive: subtract its small positive anti-Hurwitz Lyapunov solution for one direction; compare the equality and inequality solutions by the Lyapunov integral for the other. The interpolation theorem is the nontrivial sufficiency result being invoked here, not an elementary algebraic identity.

One zero, exact answer. Let $g = \frac{s-1}{s+1}$ and $h = \frac{2}{s+2}$. Then $e(1) = \frac{2}{3}$. Every stable error has $\|e\|_\infty \geq |e(1)| = \frac{2}{3}$, while

$$f = -2\frac{s+1}{3(s+2)} \quad \Longrightarrow \quad h - gf = \frac{2}{3}. \tag{87}$$

This stable proper f attains the non-strict optimum. The Pick entry is $\frac{\gamma^2 - \frac{4}{9}}{2}$, positive precisely for strict bounds $\gamma > \frac{2}{3}$. The boundary matrix is singular even though a finite optimal matching filter exists. Again, a strict feasibility test is not a theorem that the non-strict optimum is unattainable.

11 Keep the design logic in view

Signals and weights define "good." Storage certifies energy gain. Completing squares extracts feedback; duality gives estimation; coupling joins them. Finally check the full internal dynamics and the achieved norm, not just the requested certificate level.

11.1 Reproduce the examples

From the repository root, using Octave's installed `control` package:

```
octave --no-gui --quiet \
  Control/robust-control-07-hinf-synthesis/scripts/hinf_example.m
```

This checks the scalar storage example, branch/coupling boundaries, bisection, physical loop formulas, and selected exercise calculations, regenerates the figure, and checks the analytic optimum against `hinfsyn`. Norms use a continuous-system algorithm with explicit tolerance, not only a frequency grid.

Source. Scherer, chapter 7, pp. 159–216, especially §§7.3, 7.6–7.11, Theorems 7.13 and 7.40, and the eleven exercises, [1]. Examples and teaching derivations are developed here. This is not a reproduction of every technical proof; the outline labels identify the main omissions.

[1] C. W. Scherer, "Theory of Robust Control." [Online]. Available: https://www.imng.uni-stuttgart.de/mst/files/RC.pdf